Khalil El Hage Kassem

Systems Engineer & Computer Scientist

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Profile

Software engineer primarily interested in backend development and in building reliable, well-structured services that solve real business problems. Enjoys taking an idea from design to production, keeping code clean and maintainable, and picking up new tools quickly when a project needs them. Applies AI tools responsibly within a reviewed development workflow. A team player used to remote, multinational collaboration, with a self-learning mindset and clear technical communication.

Professional Experience

Software Developer Intern – *Casa Real* 02/2024 – 09/2024
Maicao, La Guajira, Colombia · On-site

- Contributed to migrating a legacy third-party on-premise ERP/CRM system to a custom cloud solution built on top of **ERPNext** and the **Frappe** framework, cutting infrastructure costs by **~20%** and subscription costs by **~30%**.
- Automated data extraction with **ETL pipelines** to securely migrate historical customer and transaction data into the new platform.

Teaching Assistant – *Computer Architectures and Algorithms & Data Structures, Universidad del Norte* 01/2022 – 12/2022
Barranquilla, Colombia

- Guided 34 students in building functional MIPS CPUs from scratch in Logisim; average group project scores rose by **23%**.
- Introduced core architecture concepts such as ISAs, pipelining, RAM and cache.
- Supported students on data structures (arrays, graphs, trees), sorting algorithms and graph algorithms (BFS, Dijkstra, A*, Floyd–Warshall), and P vs NP complexity.
- Held weekly code reviews that cut lab debugging time; exam scores improved by **15%**.

Projects

My-Blog | *Python · Django · PostgreSQL · Docker Compose · Gunicorn · Caddy* Live GitHub

- Blog platform with posts, comments and user accounts, server-side rendered with Django's MVT architecture: views pull data from PostgreSQL straight into HTML templates.
- Containerized with Docker Compose (Django app and Postgres, Gunicorn behind a Caddy reverse proxy) and running in production at blog.khalilcodes.com.

DB-Admin-WebApp | *MongoDB · Express · React · Node.js · Docker* GitHub

- Full-stack MERN app for administering a database, split into create, read, update, delete and log microservices behind an Express API gateway.
- Each service runs in its own container, so one failure leaves the rest available.

Artworks & Artists – Data Integration and Quality Pipeline | *Python · pandas · scikit-learn · ydata-profiling* GitHub

- Built an ETL pipeline that profiles, assesses and cleans the 130K-row MoMA artworks dataset using the artists dataset as a reference (transformation, error correction, deduplication).
- Enforced referential integrity across two sources by parsing multi-valued foreign keys into 13,958 artist IDs; flagged inconsistent records with IQR/z-score outlier detection and Apriori rules.

Emotion-Classifier | *Python · TensorFlow · Keras · OpenCV · Tkinter* GitHub

- End-to-end ML pipeline: data preparation, a convolutional neural network trained with TensorFlow/Keras to tell happy faces from sad ones, and TensorBoard monitoring.
- Desktop GUI in Tkinter so users can run the trained model on their own images.

3D-Briscola / *C++ · Vulkan · GLSL · CMake* GitHub

- 3D version of the Italian card game Briscola written from scratch on the Vulkan graphics API, with a playable CPU opponent, menus and keyboard controls.

Miosix Buddy Allocator – Open-Source Contribution / *C++ · Embedded OS* GitHub

- Contributed to the open-source Miosix embedded operating system by implementing a buddy memory allocator inside its kernel for 32-bit platforms, using a binary tree with both iterative and recursive variants.

Medical Appointment Mobile App / *Flutter · Dart · Firebase* GitHub

- Mobile app for booking medical appointments, backed by Firebase Auth, Cloud Firestore and Firebase Storage for authentication and real-time data.

Academic Background

MSc Computer Science and Engineering – *Politecnico di Milano* 09/2024 – 10/2026
Milan, Italy

- Coursework completed; diploma expected 12/2026.

BSc Systems Engineering (Ingeniería de Sistemas) – *Universidad del Norte* 01/2020 – 12/2024
Barranquilla, Colombia

- GPA 4.5/5. Diploma expected 12/2026; the Politecnico di Milano thesis is recognized as the final project.

Skills

Languages	Spanish — Native Arabic — Native English — C2 (TOEFL iBT 109) Italian — B1
Programming	Python · Java · JavaScript · TypeScript · SQL · C · C++ · C# · Dart · Bash
Backend	Django · FastAPI · Frappe/ERPNext · Spring · Node.js · Express · .NET · REST APIs
Frontend & Mobile	React · Tailwind CSS · HTML · CSS · Flutter
Databases	PostgreSQL · MySQL · SQL Server · MongoDB · Neo4j
DevOps & Tools	Git · GitHub · Docker · Docker Compose · Azure DevOps
Data & AI	pandas · scikit-learn · TensorFlow/Keras · LLM · Generative AI · Data Pipelines

Certificates

AZ-900: Microsoft Azure Fundamentals
DP-900: Microsoft Azure Data Fundamentals

Honours & Awards

Double Master's Degree Program, Universidad del Norte – Politecnico di Milano 2024
“Semillero de Idiomas” Scholarship (100% coverage of language courses), Universidad del Norte 2022
Highest ICFES score award, Colegio Colombo Árabe 2019